

REGION:
CROATIA

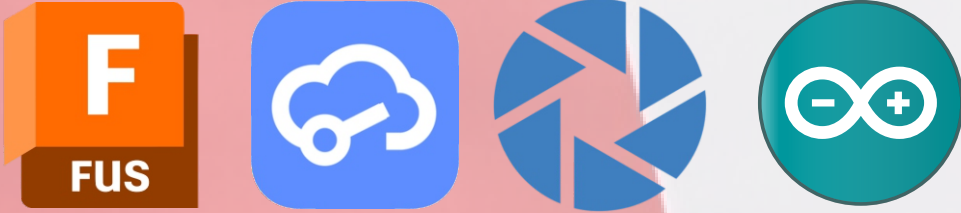


ETC VARAŽDIN
EUROPEAN TALENT CENTRE VARAŽDIN

CATEGORY:
RCJ SOCCER INFRARED

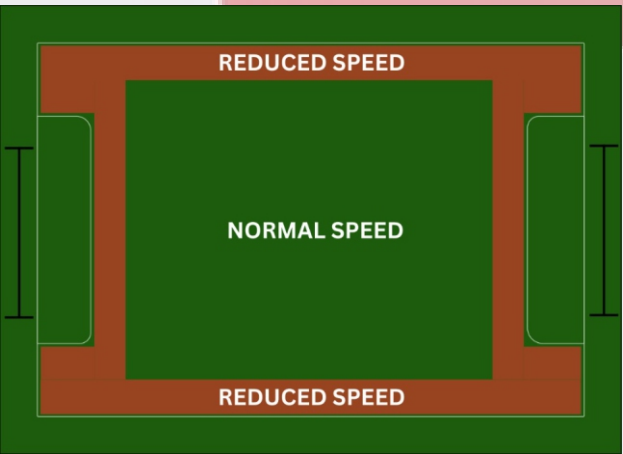
ABSTRACT

Our team from Croatia consists of two students: **Vinka Fištrek** and **David Županić**; guided by mentor **Ivan Kolarić**. Vinka led the programming, implementing core logic in Arduino C++ and camera processing in Python using the OpenMV environment as well as developing the electronical systems and designing custom PCBs in EasyEDA. David was in charge of the complete mechanical design using Autodesk Fusion360 and assembled the robot. In preparation for the competition, we developed two robots one made to play as an attacker and the other as defender. All parts were printed using the Bambu Lab X1C 3D printer located at our school and the PCBs were produced through JLCPCB. Special attention this year was given to the design of custom ball sensor to optimise ball detection precision and integration of a dribbler to extend our robot's gameplay possibilities. Both robots were developed entirely from scratch, representing a new generation of our team's work.

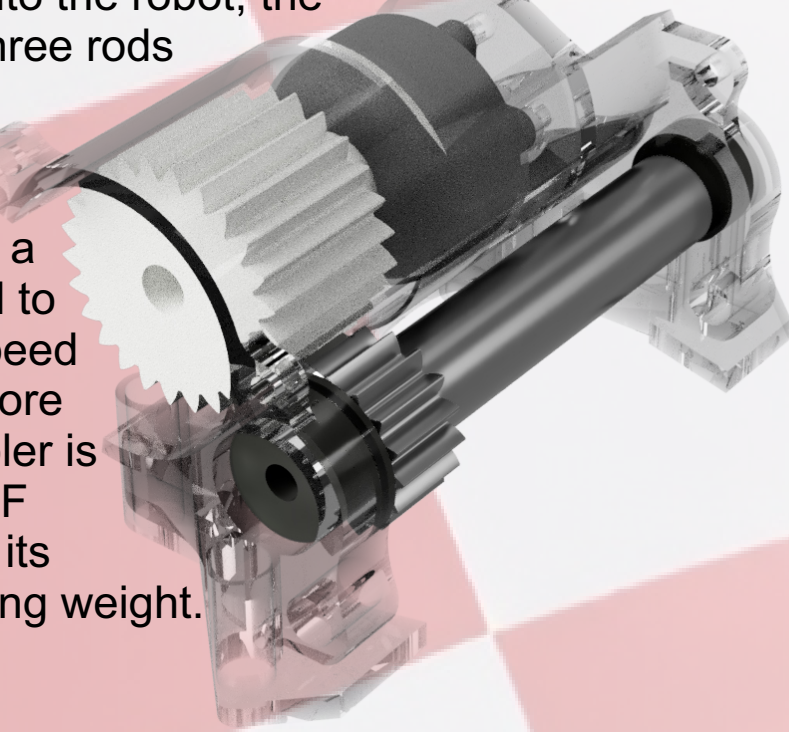


DATA, RESULTS AND DISCUSSION

Last year one of our robots' biggest flaws was the line avoidance algorithm. It was unreliable and would only work well at low movement speeds so our robots would often be too slow to be competitive. To try and fix that we added four ultrasonic distance sensors, each pointed in one of the four directions: front, back, left and right. By utilizing them to know if the robot is close to a wall, we can scale the speed of the robot so it is moving slower when next to the white lines, ensuring the white lines are correctly detected, and then speed up as it gets further away from them. To minimize the interference from the other robots we only check the distance sensors that are relative to the calculated movement angle of the robot.

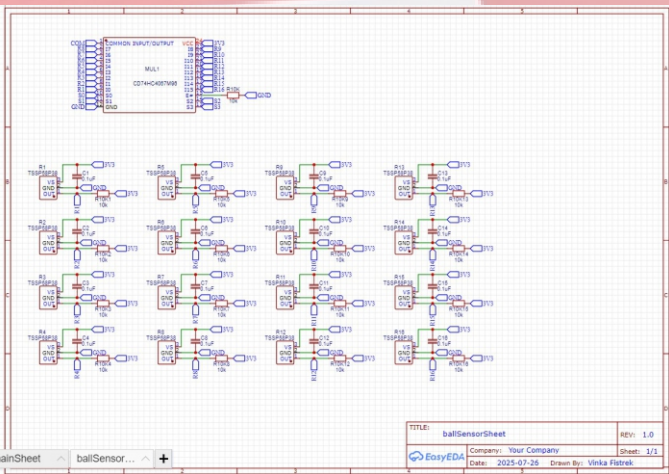


We designed the dribbler to take up as little space as possible while still performing its intended function efficiently. Instead of being mounted directly onto the robot, the dribbler is supported by three rods that allow limited vertical movement, helping it absorb impacts more effectively. The gears use a 2:1 ratio, enabling the ball to spin faster and making speed calculations easier and more accurate. The entire dribbler is 3D printed using PETG-CF filament, which increases its strength while also reducing weight.

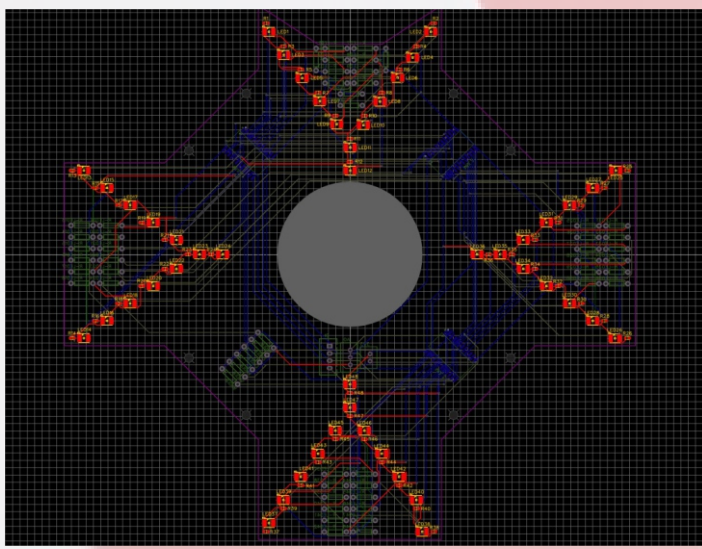


ELECTRICAL DESIGN

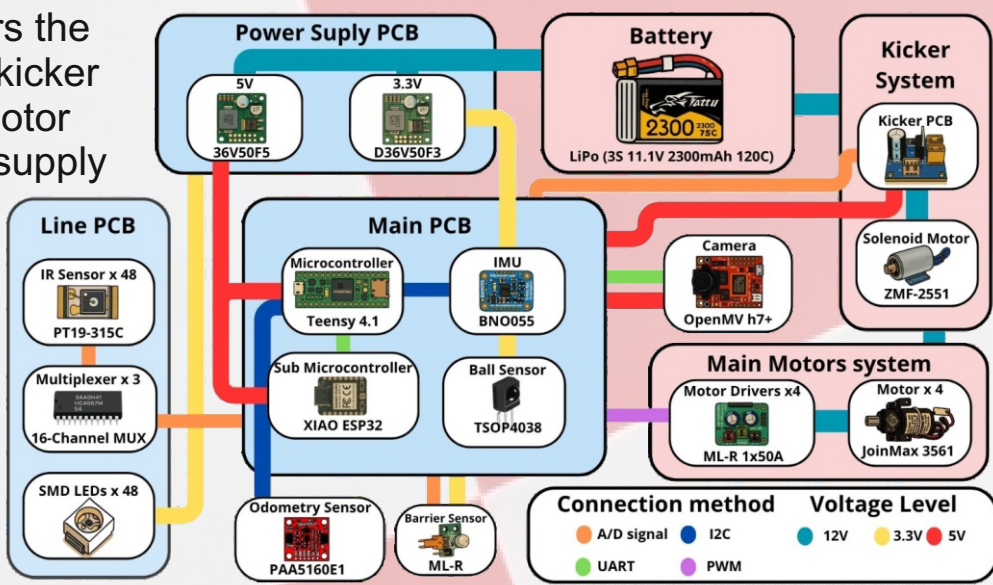
On the main PCB, the 5V rail powers the microcontrollers (Teensy 4.1 and XIAO ESP32-C6), while the 3.3V rail powers the BNO055 IMU and the integrated ball sensor. The ball sensor is connected to the ESP32, which then transmits the ball angle data to the Teensy via UART. A camera module is also connected to the main PCB; it operates at 5V and communicates with the Teensy through UART. Both the BNO055 IMU and the SparkFun OTOS are connected to the Teensy using the I2C bus. There are also four analog ultrasonic distance sensors connected to the main PCB.



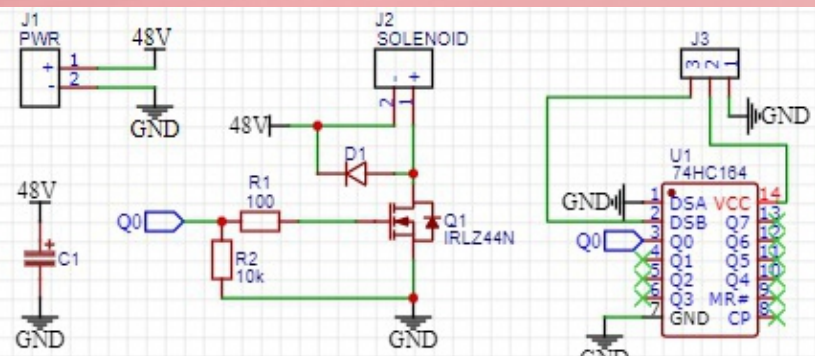
All components on the line PCB operate at 3.3V. This PCB contains 48 reflectance sensors connected to three analog multiplexers, which forward sensor data to the Teensy. Each reflectance sensor is paired with an SMD LED to improve measurement accuracy and maintain reliable operation under different lighting conditions.



Our robot's electrical system operates across three voltage levels: 12V, 5V, and 3.3V, and supports four communication protocols: analog/digital signals, I2C, UART, and PWM. A 12V battery powers the power supply PCB, kicker system, and main motor system. The power supply PCB contains two step-down voltage regulators that convert 12V to 5V and 3.3V, providing power to all other electronic components on our robots.



The kicker system is activated by a digital signal sent from the Main PCB to the Kicker PCB. Since the Main PCB operates at 3.3V logic, the signal first passes through a level shifter on the Kicker PCB, which converts it to 5V to properly drive the transistor circuit. Once triggered, the transistor allows 12V power from the battery to flow to the solenoid motor, activating the kicker mechanism. The main drive system consists of four JoinMax JMPBE-3561 motors, each controlled by an individual MRMS ML-R motor driver. The motor drivers receive PWM control signals from the Main PCB, while the motors themselves are powered directly by the 12V battery supply.



METHOD, PRODUCTION AND DESIGN

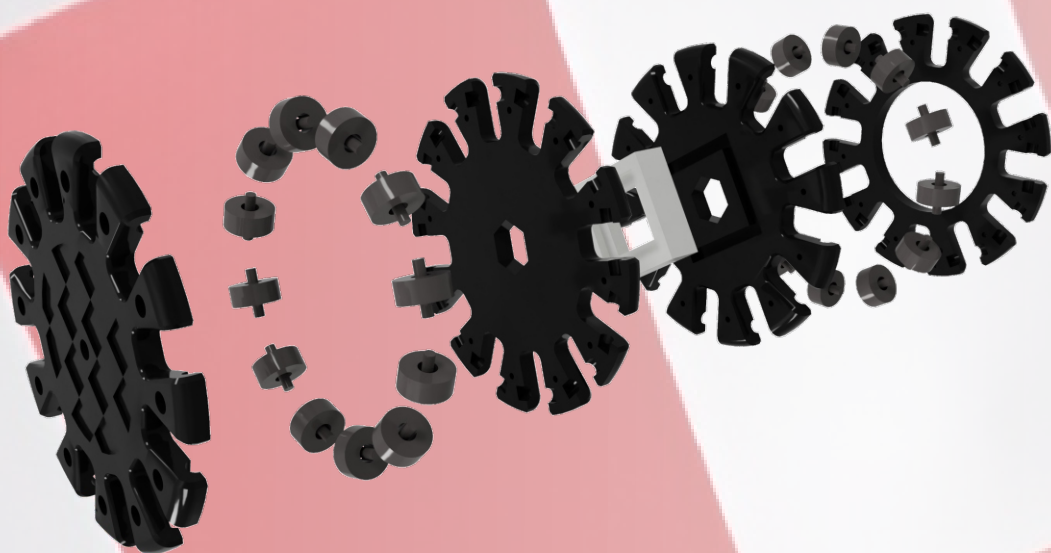
1 TEENSY 4.1
As the main microcontroller, we selected the Teensy 4.1, which we also used in last year's robot. It provides a sufficient number of input/output pins to satisfy the requirements of our system while delivering excellent computational performance for handling the robot's software and real-time processing tasks. In addition, the Teensy 4.1 includes onboard EEPROM memory, which is particularly useful for storing important persistent data such as calibrated line threshold values.

2 IMU
To maintain a stable heading, we selected the BNO055 orientation sensor, which is used to ensure that the robot moves in a straight direction and accurately rotates to a desired angle, a feature that is especially important in the goalkeeper program. During initial testing, we encountered stability issues, particularly during collisions with other robots, which occasionally caused the BNO055 to crash. To improve reliability, we implemented several hardware modifications to the PCB hosting the sensor, including the addition of capacitors and pull-up resistors on the I2C communication lines, as well as widening the PCB traces to improve signal integrity. These changes significantly enhanced the stability and overall reliability of the sensor.

3 Upper Chassis
We opted for a closed chassis design for the structure to be as compact and durable as possible. All chassis components were designed using Autodesk Fusion360, made possible through a student license granted to our team. As for the material we chose standard PLA plastic due to it being easy to work with and being sufficiently durable. It by withstanding the demands of multiple competitions without structural failure it proved to be a practical and reliable choice for our design.

4 Kicker PCB
Two of the PCB's most important components are 74LVC245 bidirectional level shifter and a MOSFET transistor. This PCB works by receiving a 3.3V signal from Teensy when the barrier sensor is triggered to the level shifter which then sends a 5V to the MOSFET which lets 12V from the battery to the solenoid. This setup lets us precisely control when the kicker system is activated, ensuring accurate and timely responses during gameplay.

5 Wheels
For this year we decided to use a dual-wheel setup which is two wheels consisting of two basic wheels. Is significantly improved both the robot's stability and made robots overall movement more precise. Additionally, since we built a new and lighter chassis, we were able to support slightly heavier wheels without compromising performance. To achieve this, we designed and ordered smaller polyurethane rubber rings from a local supplier, specifically tailored to our needs.



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10 XIAO ESP32 C6
One of the key innovations we introduced to our robots is the integration of the XIAO ESP32 microcontroller, which we use to decrease the workload on Teensy in order to lower the program runtime. This microcontroller is mounted on the main PCB and communicates with the main microcontroller (Teensy 4.1) via the UART protocol. Communication between the two robots is also made possible using the ESP-NOW protocol.

11 Main PCB
The main PCB, now in its 10th generation, serves as the core of our system. This year we went back to the circular shape of the PCB to accommodate for the integrated ball sensor. We also decided to reduce the workload on Teensy by adding the ESP32 as a second microcontroller. To ensure secure and reliable connections, we opted for JST-XH connectors.

12 Camera
Our defender robot is equipped with an OpenMV H7 Plus camera. This component plays a crucial role on the goalkeeper robot, where the camera is mounted facing backward. It enables the robot to continuously track the position of its own goal (angle of the goal centre relative to the robot and distance from the goal), enhancing spatial awareness and defensive positioning.

13 Dribbler
To have better control of the ball, we designed a dribbler. It uses an EMAX mt2204 brushless motor to spin the ball at high speeds so it constantly returns the robot and allows it to control the ball more precise than before.

6 Motors and motor drive system
Each of our robots is equipped with four Joinmax JMP-BE-3561 motors, chosen for their optimal balance of power and speed, making them well suited to meet the performance requirements of our robots. Each motor is controlled by an ML-R Motor Driver 1x50A, which features reverse voltage protection to ensure safe and reliable operation while preventing damage caused by incorrect power polarity.

7 Line PCB
This PCB contains 48 LS-PT19-315C/L177/TR8 reflectance sensors connected across 3 analog multiplexers through which Teensy can access every sensor individually. The sensors are split into four sections: front, back, right and left and the sensors are arranged in a "Y" shape since it has proven to be the best design we made so far. To improve sensor accuracy, each reflectance sensor is paired with an LED that ensures adequate illumination, allowing the robots to operate reliably even in bad lighting conditions.

8 Power supply PCB
Both robots have a power supply PCB; a TATTU 2300mAh 11.1V 75C 3S1P LiPo battery is connected to it and the PCB provides 12V, 5V and 3.3V to the entire system. This board contains two step-down (buck) voltage regulators that provide a stable 3.3V and 5V power supply to the other components.